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Hydrogen Adsorption Potential in Unconventional Carbonate Source Rocks: Challenges and Opportunities

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Abstract

The global push for decarbonization has renewed interest in underground hydrogen storage (UHS), with unconventional reservoirs emerging as potential candidates. However, the effects of source rock composition and thermal maturity on hydrogen adsorption remain poorly understood. This study utilizes the Upper Cretaceous (Maastrichtian) Jordan Source Rock (JSR) interval from the Al-Lajjun area of Western Jordan to investigate the multifaceted relationship between source rock composition, thermal maturation, and hydrogen adsorption.

The JSR is a carbonate-rich, thermally immature, Type IIS source rock that shares strong compositional similarities with key regional unconventional carbonate source rocks globally. Initial sample characterization revealed significant heterogeneity in total organic carbon (TOC) and mineralogy, including variations in carbonate and silica content. Hydrogen adsorption experiments revealed a Type I isotherm behavior, indicative of micropore-dominated adsorption mechanisms. The organic-rich, carbonate-dominated samples (i.e. JSR-1) exhibited the highest hydrogen adsorption, whereas samples with either lower TOC (i.e. JSR-3) or higher quartz content (i.e. JSR-2) showed reduced adsorption capacity. This indicates that organic matter contributes significantly to nanopore development, while mineralogical variations, particularly carbonate versus quartz content, influence the surface reactivity and available adsorption sites. Post-maturation, the hydrogen adsorption experiments revealed no detectable adsorption, likely due to the presence of in-situ generated hydrocarbons occupying pore spaces and preventing hydrogen interaction with the rock matrix. This observation highlights that thermal evolution alters not only the pore structure but also the accessibility of adsorption sites.

Collectively, the findings underscore the importance of both organic richness and mineral matrix in controlling hydrogen adsorption in immature source rocks. However, as maturity progresses and hydrocarbons are generated, hydrogen adsorption potential diminishes regardless of initial composition. This suggests that thermally mature, depleted source rocks with limited residual adsorption potential may offer favorable conditions for cyclic or seasonal hydrogen storage, especially where retrieval efficiency is critical.

Introduction

The global shift toward low-carbon energy systems has intensified interest in hydrogen as a sustainable, flexible, and zero-emission energy carrier. As countries commit to net-zero targets, hydrogen is increasingly seen as a cornerstone of future energy transitions, with applications spanning power generation, transportation, industrial processes, and long-duration energy storage [1, 2]. To meet the anticipated demand for large-scale hydrogen infrastructure, robust and efficient storage technologies are essential, particularly for balancing supply-demand fluctuations, integrating renewable energy sources, and ensuring energy security [3–5].

Current hydrogen storage technologies include compressed gas, cryogenic liquid, and solid-state storage in metal hydrides or porous materials. However, these methods face substantial technical and economic barriers, such as high energy input for compression or liquefaction, low volumetric density, material degradation, and limited scalability [6–8]. In this context, subsurface geological storage has emerged as a promising option for bulk hydrogen storage, leveraging existing knowledge from natural gas storage and CO₂ sequestration [9, 10].

Underground hydrogen storage (UHS) in salt caverns is already technically viable and operational in several regions. However, the global distribution of suitable salt deposits is geographically limited, prompting interest in alternative geological formations such as depleted oil and gas fields, deep aquifers, and organic-rich unconventional source rocks [9, 11]. Among these, source rocks and other unconventional formations remain underexplored, especially regarding their potential to adsorb and retain hydrogen under subsurface conditions.

Source rocks, typically rich in organic matter and fine-grained mineral matrices, exhibit nanoporous structures and reactive surfaces that can offer significant storage potential, particularly through adsorption-based mechanisms [12]. However, the source rocks are compositionally heterogeneous (figure 1), and the hydrogen storage capacity of such rocks is likely influenced by their organic and inorganic geochemical composition, thermal maturity, pore structure, and existing hydrocarbon content [13, 14]. These factors introduce significant complexity and uncertainty, which must be understood to evaluate their storage feasibility.

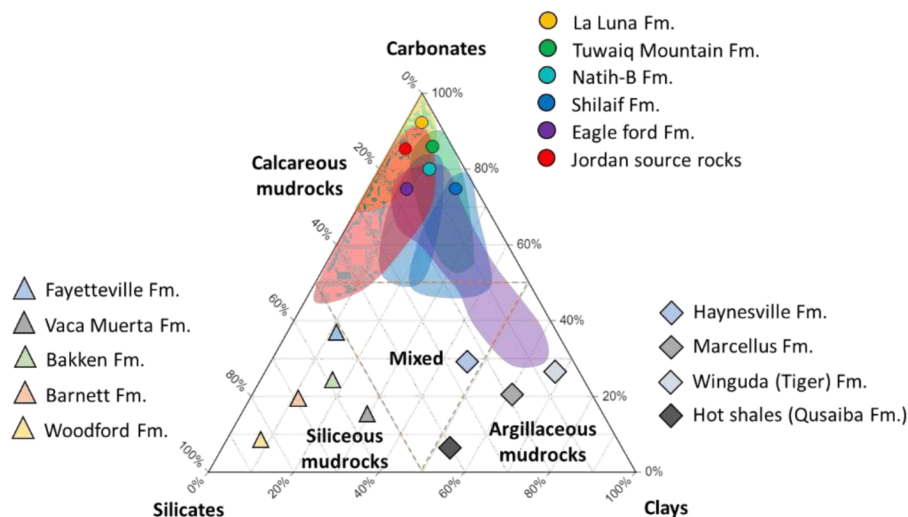


Figure 1—Mineralogy ternary diagram representing source rocks of different origin and composition. The shaded color represents the compositional variability associated with carbonate source rocks [15]

The primary aim of this study is to investigate the influence of geochemical composition and thermal maturity on hydrogen adsorption in organic-rich source rocks, using the Maastrichtian Jordan Source Rock (JSR) as a case study. Through a multidisciplinary approach combining detailed geochemical analysis,

artificial maturation experiments, and hydrogen adsorption measurements, we examine the following central research question "How do variations in organic richness, mineral composition, and thermal maturity affect the hydrogen adsorption capacity of carbonate-rich source rocks?"

This study also seeks to determine whether matured, previously hydrocarbon-generating source rocks are still viable targets for hydrogen storage, or if hydrogen adsorption is effectively inhibited due to the presence of residual hydrocarbons. Understanding the controls on hydrogen adsorption in source rocks is critical for identifying viable storage candidates in regions lacking salt formations. It also offers new perspectives on how petroleum basins could be repurposed for energy storage in a hydrogen-based economy. The findings from this study provide important technical insights that support ongoing efforts to assess and deploy seasonal or strategic hydrogen storage in unconventional reservoirs.

Methodology

To evaluate the influence of compositional heterogeneity on hydrogen adsorption, core plug samples representing varying organic matter content and mineralogical composition were selected from a well drilled in Jordan source rock (JSR) interval in central Jordan (Fig. 2). The JSR was selected due to its high total organic carbon (TOC) content and prominent compositional heterogeneities, which are comparable to those observed in many globally significant unconventional petroleum source rocks (e.g., the Tuwaiq Mountain and Jubaila Fm in Saudi Arabia, La Luna Fm in Venezuela, Eagle Ford Fm in the USA, Vaca Muerta Fm in Argentina, and Shilaif Fm in UAE) [16–21]. The detailed stratigraphic and geochemical framework of the drilled well, highlighting compositional variability and geochemical cyclicity in JSR, is discussed in Usman et al. (2025)[22]. Furthermore, the thermally immature nature of the JSR makes it particularly suitable for controlled laboratory experimentation, enabling us to assess the effects of artificial maturation on hydrogen adsorption. A multi-stage workflow was employed including bulk organic geochemical and mineralogical characterization, artificial maturation via hydrous pyrolysis, and hydrogen gas adsorption measurements on both immature and matured samples.

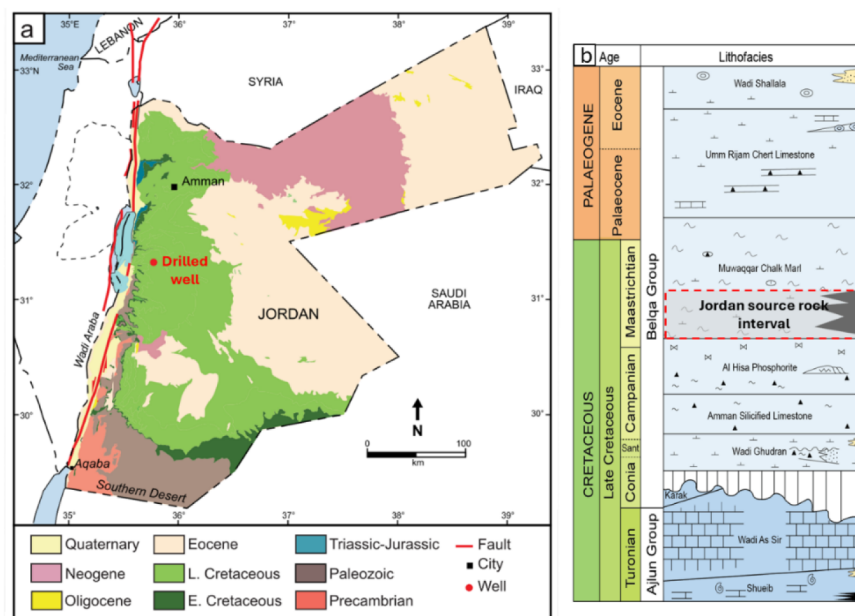


Figure 2—(a) Geological map of Jordan showing the drilled well location [22]. (b) Stratigraphic column showing the deposition of carbonate dominated strata along with Jordan source rock interval during the Cretaceous time period [23].

Sample characterization and artificial maturation

The bulk organic geochemical properties of the selected samples were determined using a Rock-Eval 7S pyrolysis system. The parameters obtained include total organic carbon (TOC, wt%), free hydrocarbons (S1, mg HC/g rock), thermally cracked hydrocarbons (S2, mg HC/g rock), thermal maturity indicator (Tmax, °C), hydrogen index (HI, mg HC/g TOC), oxygen index (OI mg CO/g TOC), mineral carbon (MinC, wt, %), and total sulfur (TS, wt %). The analytical procedure follows the guidelines and standards established by Behar et al. (2001) and H.C. Ortiz et al. (2021) [24, 25]. An IFPEN-160 K standard was routinely used before and after each analytical run to ensure precision and reproducibility.

Based on the results of the initial characterization, a representative plug sample exhibiting high TOC and elevated carbonate content was selected for artificial maturation. Hydrous pyrolysis experiment was performed at 330°C to simulate burial-induced thermal maturation of the source rock sample. The experimental procedures for hydrous pyrolysis followed the protocols outlined by Usman et al. (2024) [15]. Following maturation, the sample was re-analyzed using Rock-Eval 7S to document any changes in geochemical parameters resulting from thermal transformation.

Hydrogen Adsorption

Hydrogen gas adsorption experiments were performed using an automated Sievert's-type PCT Pro instrument, allowing for precise control of pressure and temperature conditions. The experimental setup used high-purity hydrogen gas (H₂, 99.99 mol%) as the adsorbate, with helium (He, 99.99 mol%) employed as the calibration gas. Instrument-grade air (SG) was used to operate the pneumatic system and maintain stable functionality throughout the measurements. Prior to each experiment, the rock samples were oven-dried and degassed under vacuum for 12 hours to eliminate residual moisture and pre-adsorbed gases from the pore system. Subsequently, approximately 0.5 to 1.0 grams of each powdered sample was placed in the instrument's sample holder. The sample chamber was evacuated at 30°C for 60 minutes to ensure baseline conditions. Adsorption experiments were carried out isothermally at 30°C. Hydrogen gas was introduced incrementally into the chamber under controlled pressure steps, and the system continuously recorded pressure-volume changes to monitor adsorption behavior in real time. Adsorption isotherms were generated by varying the pressure systematically, and the resulting data were used to calculate the hydrogen adsorption capacity of each sample. These experiments were conducted on both the original (immature) and thermally matured samples to evaluate the influence of maturation and compositional changes on adsorption behavior.

Results & Discussion

The bulk geochemical characterization of the JSR samples reveals marked variability both in organic matter content and mineralogical composition. The total organic carbon (TOC) varies significantly between the samples where JSR-1 and JSR-2 shows high TOC (19.1 and 19.5 Wt. %) where as JSR-3 shows relatively lower TOC (7.7 wt. %). The samples also showed relatively high hydrogen Index corresponding to a Type II kerogen (Figure 3). However, all the three samples have lower Tmax, showing an immature source rock interval. With maturation of JSR-1 sample at 330°C, the Tmax increased significantly, now representative of a mature source rock interval in peak oil generation window (Figure 3).

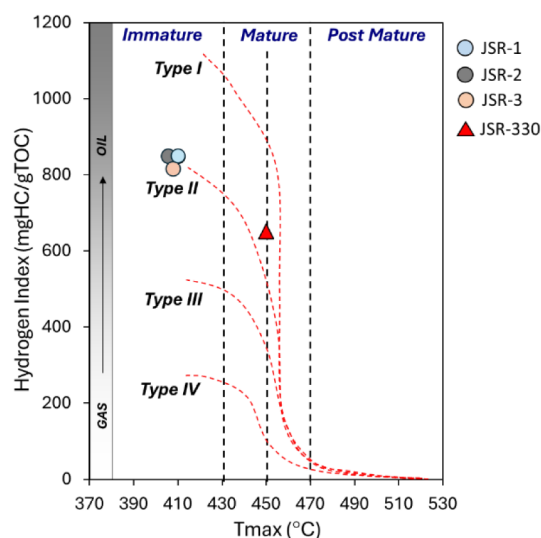


Figure 3—Bulk organic geochemical results showing kerogen type and thermal maturity of the samples before and after artificial maturation

The analyzed samples also showed mineralogical variability, further enhancing the interpretation of compositional controls on hydrogen adsorption behavior in unconventional reservoirs. The XRD results indicate that JSR-1 and JSR-3 are carbonate-dominated, with calcite as the dominant mineral phase. In contrast, JSR-2 represents higher quartz content, calcite, and other minor mineral phases in the sample. This contrast is critical to understand the effect of mineral matrix composition along with the presence of organics that can significantly affects surface area, pore size distribution, and ultimately gas adsorption characteristics in unconventional reservoirs [26, 27].

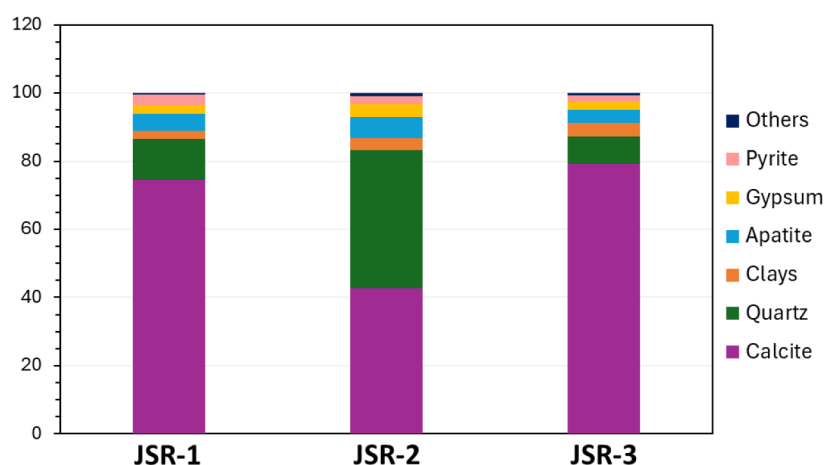


Figure 4—XRD results showing mineralogical composition of the analyzed samples

Effect of source rock organic and inorganic composition on hydrogen adsorption

Hydrogen adsorption experiments conducted on immature JSR samples reveal that all the tested samples exhibit Type I adsorption isotherms as classified by IUPAC classification [28]. The results indicated a steep initial increase in adsorption at low relative pressures, followed by a plateau as equilibrium is reached (Figure 5). This type of isotherm is indicative of monolayer adsorption on microporous solids (pores <2 nm), which is consistent with the dominance of nanoporous networks within organic-rich mudrocks [29, 30]. In the context of hydrogen, the high initial uptake is likely due to its preferential filling of micropores that provide high-energy adsorption sites. Once these primary sites are saturated, further adsorption is minimal, reflecting the monolayer capacity of the matrix [28, 31].

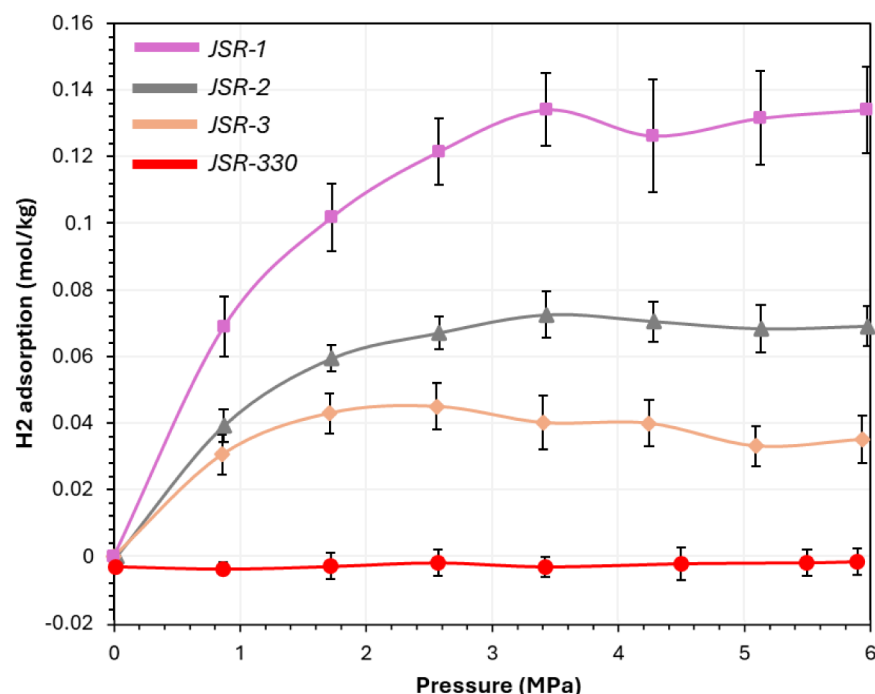


Figure 5—Hydrogen adsorption results of the immature and artificially matured JSR samples with error bars.

The experimental results clearly show a positive correlation between TOC and hydrogen adsorption capacity (Figure 5). The carbonate-rich, high-TOC sample (JSR-1) demonstrates the highest adsorption capacity compared to JSR-3, which is also carbonate-dominated but contains substantially lower organic content. This relationship supports the prevailing understanding that organic matter is a principal contributor to gas storage in shale and mudrock systems, not only through its inherent nanopore networks but also via surface interactions [31, 32]. Kerogen in its immature state contains abundant functional groups (e.g., aromatic, carboxylic, and hydroxyl) that may contribute to enhanced physisorption of small gas molecules like hydrogen due to favorable van der Waals interactions and dipole-induced effects [33, 34].

However, the influence of mineralogical composition also becomes apparent when comparing samples with similar TOC but different mineral matrices. For instance, JSR-2, which is quartz-rich but contains TOC comparable to JSR-1, shows noticeably lower hydrogen uptake (Figure 5). This indicates that the mineral matrix plays a modifying role in hydrogen adsorption beyond TOC alone. While both calcite (CaCO_3) and quartz (SiO_2) are considered relatively inert with low intrinsic gas adsorption capacities, their interaction with organics and structural properties differ [35, 36]. Calcite, with its ionic lattice and polarizable surface sites, may facilitate stronger adsorption through induced dipole interactions, particularly in the presence of polar organic compounds. In contrast, quartz surfaces are dominated by silanol (Si-OH) groups, which are less chemically interactive in dry systems and offer lower surface energies for hydrogen sorption [33, 37]. Additionally, the mineral-organic interactions may also affect pore structure development and connectivity [38, 39]. The observed behavior in JSR-2 may therefore stem from a lower effective surface area and a different distribution of active sites for hydrogen sorption.

These findings emphasize that immature, organic-rich source rocks are capable of retaining significant volumes of hydrogen, particularly in settings dominated by carbonate matrices. However, such strong adsorption may also pose a challenge for gas recoverability, as hydrogen exhibits relatively high affinity to polar organic functional groups, potentially leading to lower desorption efficiency and reduced retrieval during production phases [30, 40]. This has direct implications for subsurface hydrogen storage in such lithologies, where injection feasibility may be high, but retrievability and cyclic deliverability may be limited by strong physisorptive interactions within the organic framework.

Effect of source rock maturation on hydrogen adsorption

To assess the impact of thermal maturation on hydrogen adsorption behavior, the high-TOC, carbonate-dominated source rock sample (JSR-1) was artificially matured under controlled laboratory pyrolysis conditions to simulate post-oil window thermal maturity. The results indicate a complete absence of hydrogen adsorption in the artificially matured sample (Figure 5), in contrast to its immature counterpart which showed Type I isotherm behavior. This sharp decline in adsorption capacity upon maturation provides important insights into the role of organic matter transformation and hydrocarbon presence in governing adsorption dynamics.

The absence of hydrogen uptake in the mature sample can be primarily attributed to the presence of remaining hydrocarbons within the pore network during pyrolysis. Thermal maturation leads to the transformation of kerogen into bitumen and hydrocarbons, many of which remain trapped within the matrix due to poor pore-network connectivity [41-43]. The produced in-situ hydrocarbons during maturation will therefore block the pore spaces and adsorption sites, thus impeding the ability of hydrogen molecules to interact with the rock surface. Similar observations have also been observed in prior studies on gas sorption in hydrocarbon-saturated shales [32, 35, 44-46].

Moreover, the thermal transformation of organic matter leads to the loss of polar functional groups (e.g., hydroxyl, carboxyl, and carbonyl groups) that are abundant in immature kerogen and contribute to favorable van der Waals and induced dipole interactions with small gas molecules such as H_2 [26, 46, 47]. As maturity increases, the kerogen structure becomes increasingly aromatic and inert, reducing its surface reactivity and adsorption potential.

These observations suggest that thermally mature source rocks, particularly those that are overmature or have passed through the main hydrocarbon generation window, may serve as more suitable candidates for subsurface hydrogen storage. In such systems, the reduced or negligible adsorption capacity implies that hydrogen would experience minimal physisorption losses, enhancing the retrievability and cyclic deliverability of injected gas. However, the presence of residual hydrocarbons within pore spaces may still reduce effective pore volume available for hydrogen storage and may complicate the predictability of storage capacity. Therefore, for optimal hydrogen storage design in unconventional reservoirs, reservoirs with high maturity but depleted hydrocarbon content, such as post-mature source rocks or shales may provide a favorable balance between minimal hydrogen adsorption and sufficient storage space.

Conclusion

This study provides new insights into the hydrogen adsorption behavior of immature and artificially matured organic-rich source rocks, highlighting the crucial role of both organic and inorganic composition as well as thermal maturity. The immature samples exhibited Type I isotherm behavior, indicative of microporous adsorption dominated by nanopore networks. Hydrogen uptake was found to correlate positively with total organic carbon (TOC) content, underscoring the importance of organic matter in creating adsorption-favorable pore structures. Among samples with similar TOC levels, mineralogical variations, particularly between carbonate and quartz-dominated matrices, also influenced adsorption capacity, with carbonate-dominated samples displaying enhanced hydrogen uptake, potentially due to higher surface reactivity and active site availability.

In contrast, artificially matured samples showed no hydrogen adsorption due to the occlusion of pore spaces by in-situ generated hydrocarbons and the transformation of reactive functional groups into more inert aromatic structures. These findings suggest that thermally mature, hydrocarbon-depleted source rocks may serve as more favorable candidates for subsurface hydrogen storage due to their minimal hydrogen adsorption and potentially higher recovery efficiency.

Overall, the results emphasize the need for careful compositional and maturity screening in evaluating unconventional reservoirs for hydrogen storage. The dual influence of organic matter content and mineral

matrix composition must be integrated into reservoir screening protocols to optimize both storage capacity and retrieval efficiency. Further research, including high-pressure cyclic adsorption experiments, samples covering different ranges of thermal maturity, and long-term hydrogen exposure studies, is warranted to better understand the stability and performance of these systems under realistic subsurface conditions.

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